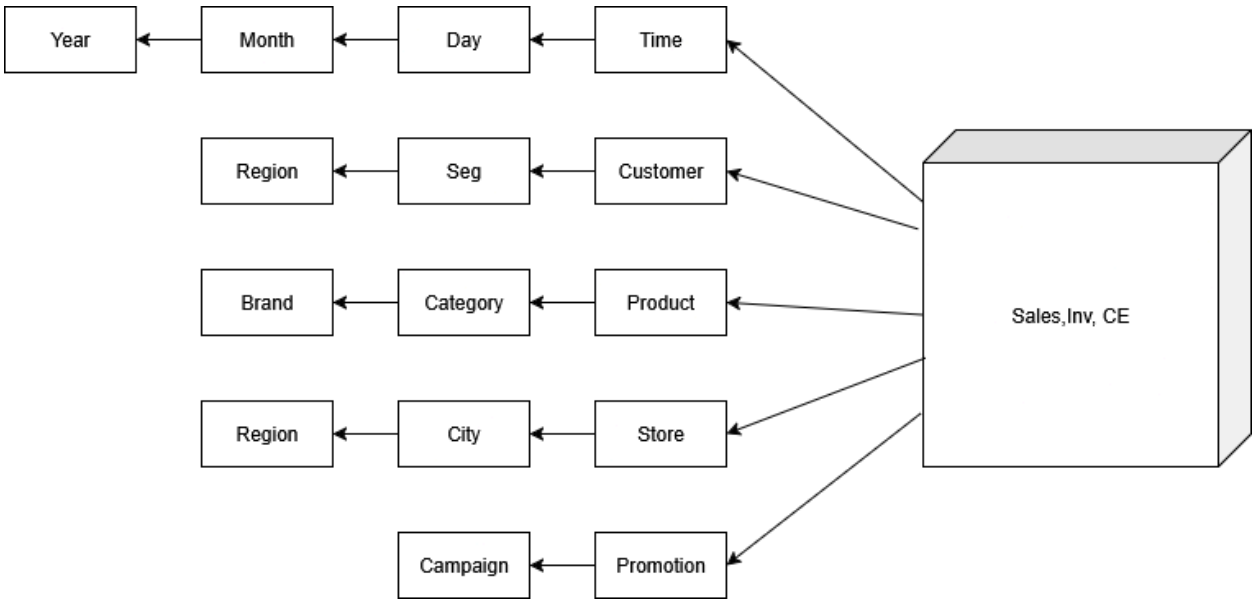
Q1(1)



Q1(2) EcoTrend should use a **hybrid approach**:

- **RDB (Relational Database)** for storing raw, detailed transactional data (sales, inventory, customer records). It's efficient for updates, consistency, and integration from multiple sources.

- **MDB (Multidimensional Database)** for analytical queries and OLAP (slicing, dicing, roll-ups). This supports fast performance on trend analysis, market insights, and inventory turnover checks.
  Together, this ensures both reliable data storage and high-speed analysis.

Q1(3) **Important Dimensions & Facts**

Dimensions: Product, Customer, Store, Time, Promotion.

Facts: Sales (amount, quantity, discounts), Inventory (levels, turnover), Customer Engagement (visits, purchases, loyalty points).

Insights:

- Identify slow-moving products and optimize stock.

- Spot loyal vs. one-time customers and target promotions accordingly.

- Compare e-commerce vs. store performance across regions.
  This helps EcoTrend improve turnover, reduce stockouts, and build stronger customer retention strategies.

Q1(4) **ETL Process**

- **Extraction:** Pull data from POS systems in 15 stores and the e-commerce platform (sales, inventory, customer activity).

- **Transformation:** Standardize formats (e.g., date formats, product codes), clean duplicates, calculate derived metrics (turnover rate, profit margins), and ensure consistent naming conventions.

- **Loading:** Store cleaned data in the central warehouse. Detailed data into the RDB layer; aggregated summaries into the MDB for OLAP.
  This ensures a single, consistent view of business operations.

Q2: **Data Warehouse Design**

- Use a **centralized global data warehouse** with **regional data marts**.
    - Central DW = consolidated, consistent reporting across all regions.
    - Regional marts = handle local regulatory, currency, and compliance needs.
- Storage Approach: Hybrid (RDB for detailed data, MDB/OLAP cubes for fast analysis).

Regional Differences in Storage

- Currency formats (USD, EUR, PKR, etc.)
- Regulatory requirements (GDPR in Europe, data localization in Asia)
- Taxation and pricing structures
- Time zones and date formats
- Supply chain tracking granularity (depends on local vendors and logistics rules)

Data Warehouse Operations

- **Best Collection Method:** ETL pipelines with automated connectors (APIs, ERP/CRM integration, IoT feeds from manufacturing). Ensures consistency, standardization, and scalability.

Five Problems & Solutions in Combining Data:

1. **Inconsistent formats** → Standardize units, dates, and codes.
2. **Duplicate records** → Deduplication rules and unique identifiers.
3. **Missing data** → Use imputation or business rules for defaults.
4. **Different data quality levels** → Validation checks and cleansing.
5. **Regulatory conflicts** → Apply region-specific compliance filters before integration.

Bulk Loading Time

- Best during **off-peak business hours** (e.g., late night/early morning in each region). Minimizes system slowdown and avoids interfering with daily transactions.

Business Analysis – Five Key Questions:

1. Which products are performing best in each region?
2. How do supply chain delays differ across continents?
3. What is the ROI of R&D investments by region?
4. Which customer segments show the highest loyalty?
5. How do regulatory differences impact product launch timelines and sales?

Q3:**1. Relational Database (RDB) for Transactional Data**

- Store detailed **row-level transactions** in normalized tables. Ensures data integrity and supports frequent inserts/updates.
- Example **Sales Table:**
-

```
Sales(
    SaleID INT PRIMARY KEY,
    CustomerID INT,
    ProductID INT,
    Quantity INT,
    Price DECIMAL(10,2),
    SaleDate DATE,
    StoreID INT
)
SELECT CustomerID, SUM(Quantity * Price) AS TotalSales
FROM Sales
GROUP BY CustomerID;
```

2. Multidimensional Database (MDB) for Trends

- Organize data into a cube: Facts (measures) at the center, Dimensions around it.
- Cube Structure Example:
    - Fact: Sales Amount, Quantity
    - Dimensions:
        - Time (Day → Month → Year)
        - Product (Item → Category)
        - Region (Store → City → Country)
        - Customer (Customer → Segment)
- **OLAP Operations Supported:**
    - Slice: Filter on one dimension (e.g., sales in 2025 only).

- Dice: Select subcubes with multiple filters (e.g., smartphones in Europe, Q1 2025).
- Roll-up: Aggregate up a hierarchy (Day → Month → Year).
- Drill-down: Go from high-level (Yearly sales) to detail (Daily sales).

3. MDB vs RDB

- Advantage of MDB: Faster analysis for aggregated queries (OLAP cubes are optimized for slicing/dicing large datasets).
- Disadvantage of MDB: Poor at handling frequent updates and detailed transactions compared to RDB (less flexible for real-time transactional data).

Q4:

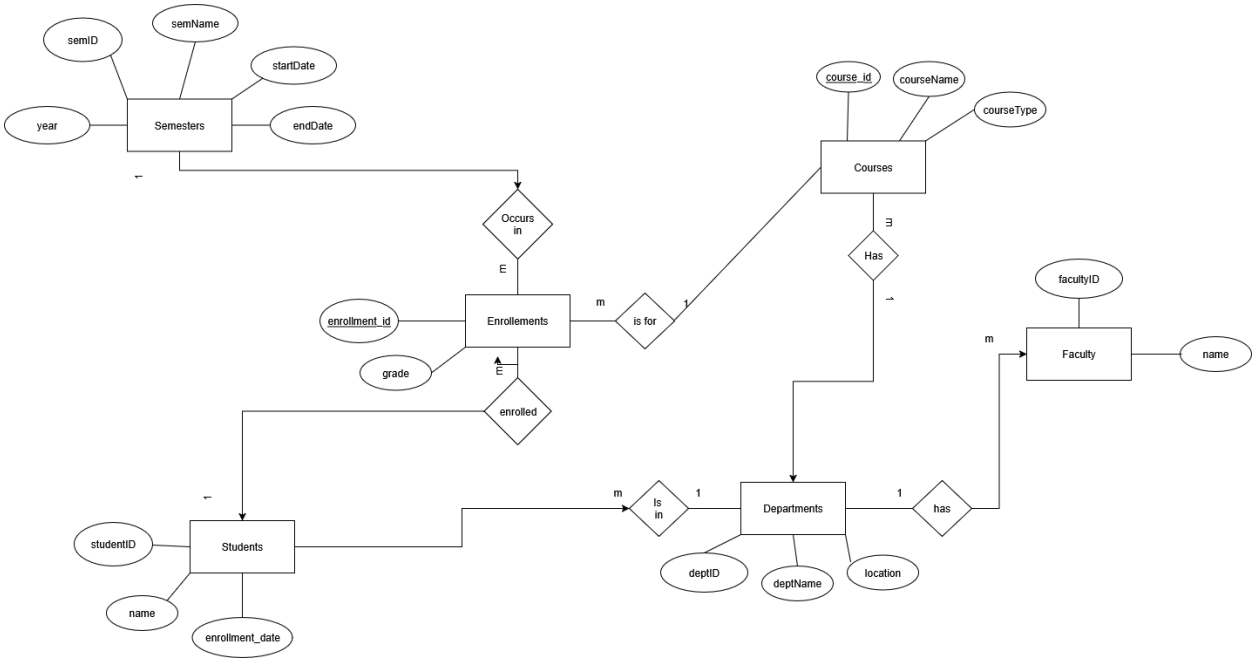
1. ERD (Entity Relationship Diagram)

Entities and relationships:

- **Students** (student_id PK, name, enrollment_date, department_id FK)
- **Courses** (course_id PK, course_name, course_type, department_id FK)
- **Enrollments** (enrollment_id PK, student_id FK, course_id FK, grade, semester_id FK)
- **Semesters** (semester_id PK, semester_name, start_date, end_date, year)
- **Departments** (department_id PK, dept_name, location)
- **Faculty** (faculty_id PK, name, department_id FK)

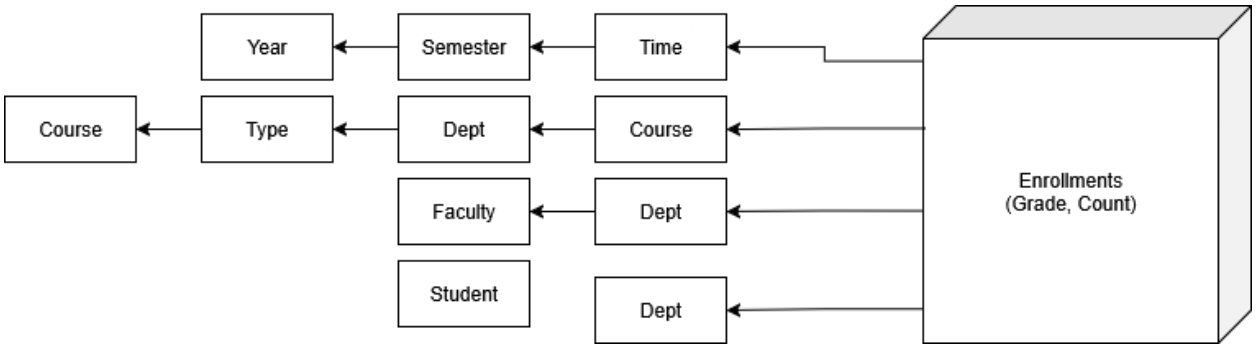
Relationships:

- Students ↔ Courses = Many-to-Many via Enrollments
- Departments → Courses (1:M)
- Departments → Faculty (1:M)
- Semesters → Enrollments (1:M)



2. MER (Multi-Dimensional Model)

- **Fact Table:** Enrollments
    - Measures: Grade, Enrollment Count
- **Dimensions & Hierarchies:**
    - **Time:** Year → Semester
    - **Course:** Department → Course Type → Course
    - **Student:** Department → Student
    - **Faculty:** Department → Faculty



OLAP Operations:

- **Roll-up:** Aggregate grades by Year instead of Semester.
- **Slice:** Analyze only “Computer Science” department enrollments.
- **Dice:** Select subcube (e.g., 2024 Year + “Business Dept” + “Elective Courses”).

3. **Short Note (ERD → MER Conversion)**
The ERD captures how the university operates—entities like Students, Courses, Semesters, and their relationships. However, for analytics we transform this into an MER model, where the central focus is the fact table (Enrollments) that stores measurable events. The surrounding entities become dimensions that provide context: Time, Course, Student, and Faculty. Hierarchies inside dimensions (e.g., Year → Semester, Department → Course) allow OLAP operations such as roll-up, drill-down, slice, and dice. This conversion ensures the data warehouse supports both detailed academic reporting and higher-level trend analysis.

Q5:

1. Geographically Distributed Data Warehouse

- **Definition:** Data is spread across multiple regions (e.g., NA, EU, Asia) close to users and regulators.

a) **Benefits:**

- Meets **data sovereignty laws** (GDPR in EU, data localization in Asia).
- Reduces **query latency** for regional users by storing data locally.
  Challenge: Synchronization delays between regions → risk of inconsistent global

views.

b) **Example:**

- Store EU customer data in AWS Frankfurt (GDPR compliance), NA data in AWS Virginia, Asia data in Singapore.
- Use **data replication pipelines** (Kafka, Snowflake replication, BigQuery cross-region) to sync aggregated data into a central global warehouse for exec dashboards.

2. Technologically Distributed Data Warehouse

- **Definition:** Uses multiple technologies (cloud + on-prem + specialized tools) working together as one logical warehouse.

a) **Benefits:**

- Integrates **legacy on-prem systems** with modern cloud platforms (no need to throw away old infra).
- Enables workload specialization (e.g., OLAP cubes in cloud, ETL staging on-prem).
  Challenge: Complexity of integrating heterogeneous systems, with possible vendor lock-in.

b) **Example:**

- On-prem Oracle DB for legacy inventory logs.
- Snowflake or BigQuery in cloud for analytics.
- **ETL/ELT pipelines** (Informatica, Airflow, Fivetran) sync data from both into a unified warehouse.

3. Hybrid Model (Geographic + Technological)

- Combine **geo-distribution** for sovereignty + **tech-distribution** for tool integration.
- Example:

    - Regional warehouses: EU → AWS Frankfurt (GDPR), NA → Snowflake US, Asia → GCP Singapore.

    - Tech integration: Cloud analytics + legacy ERP on-prem, stitched via ETL pipelines.

- **Trade-offs:**

    - **Consistency:** Eventual consistency across regions; may sacrifice real-time global view.

    - **Cost:** Higher infra + replication costs, but compliance and speed are worth it.

    - **Scalability:** Strong; can scale cloud nodes while keeping local compliance.